MCV 4U1 Trigonometric Worksheet

1. Determine $\frac{dy}{dx}$ for each of the following.

a. $y = \sin(4x) + \tan^2(x)$

b. $y = \cos^3\left(x^3\right) + \tan\left(\frac{\pi}{4}\right)$

c. $y = \sec\left(\sec\left(2x\right)\right)$

d. $y = \sin^3(\cos(x)) + \sec(\sqrt{x} + 1) - \frac{4}{\csc(2)}$

2. Determine the coordinates of the point(s) where the tangent line to the curve $y = 2\sin(x) - \cos(2x)$ is horizontal over the interval $x \in \left[\pi, 2\pi\right]$.

3. Determine the equation (in point slope form) of the normal line to the curve $y = -3\cos(x) + 1$ when $x = \frac{\pi}{6}$.